

John Atkinson philosophies in preparing swimmers and in coaching

- The Age Group athlete through frequency of exposure to the training stimulus and some competition must be taught and developed so that there is the highest possible level of "skill" development and enhancement.
- Slow aerobic swimming where there is high skill demand and technique 'quality control on the Age Group athlete in developing superior technique and skills', is far more important as a "technical" base in preference to the much emphasised "aerobic" base.

LTS.....

- The "LTS" philosophy
- "LTS" is Learn To Swim,
- Love The Sport,
- and Learn The Skills.
- These must be part of all Age Group Development Programmes.

- Swim and mini-squad programme. All coaches must be teachers, and all teachers should be coaches.
- It's the attention to detail by the teacher that makes the difference. Communication is the key factor with the great Learn to Swim teacher.

- Correct skill acquisition of swimming for distance is a requirement that should be reinforced prior to any new skill being introduced.
- Speed and endurance can be delivered to skill but rarely if ever does the reverse apply.

- It is also worthy to note that as against front crawl which has 80% of flawed skill, backstroke has been the No.1 stroke in the early developing years of a significant majority of the world's best swimmers. Arguably, there is a great need to consider backstroke as the core stroke in a young athlete's career and it could be taught higher up the learn to swim chain earlier.

Rules/Guidelines of Biomechanics

- Head down – hips up (minimise resistance)
- All actions are slow into fast (and applies to learning process)
- Feel the water – you cannot feel what you can't touch (i.e. sculling etc.)
- Fingers first – applied force is initiated and held by leading with fingers first
- FS elbow led recovery, after full extension, not hand, until a certain point

A coach

- Offers the Club Board/Committee professional advice regarding the overall direction of the club especially regarding team goals and training objectives, team competitions and club selections.
- Provide a positive, progressive goal orientated teaching, development and training programme to meet the needs of the club's competition schedule.
- Provide coaching and advice to challenge talented individuals within the club and to encourage increased participation, involvement and commitment.
- To teach, coach and train the club's more advanced and senior dedicated and committed athletes. Provide quality coaching and quality time with quality athletes.
- To educate, advise and direct the coaching staff in identification, education, motivation, participation and evaluation of the club's athletes and parents. Provide a clearly identifiable teaching, coaching and training model to suit, stimulate and challenge the athlete's needs.

A Head Coach

- The Head coach would oversee and supervise this programme without interference or compromise from the Club Board or parents within the club on acceptance of the plan and programme by the Club's Board or parent body.
- To ensure the club has adequate, appropriate and suitable pool space, gym equipment, training equipment etc by communicating these needs to the club board or by providing or arranging for provision of these items.
- With the coaches' capabilities, ensure the health and well being of swimmer participating in the programme. This includes injury prevention skills and drug education skills.
- The club and Head Coach should work together to build a fully integrated, multi dimensional teaching, training and education programme.
- The club must accept and understand that it is not possible nor is it expected that the Head Coach can coach each swimmer in the club. Whilst this may be desirable by both parents and Head Coach, it is not practical or possible to suggest it. The Club has the Head Coach's programme, not the Head Coach coaching each and every swimmer in the programme. Allowance must also be made for periods when the Head Coach is absent at competitions/camps with the senior or higher achieving athletes.

- When establishing club programmes that are 'new' it is an opportunity to build something how the head coach would like it right from the start.
 - Think outside the box.
 - You have an opportunity to build a fresh rather than going with the old ways.
- The Question I would ask you as the head coach is,
 - Should you go with any swimmers who want to join your new programme or put an age limit on who can join?

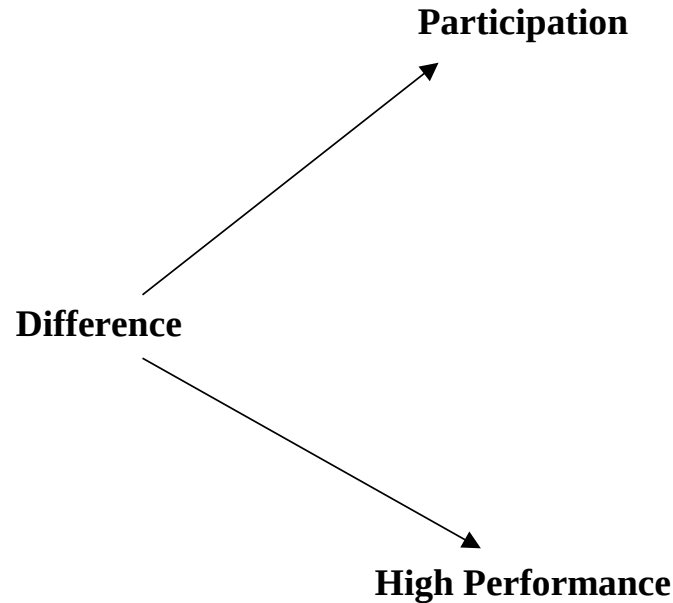
- When you establish a new programme some people in other clubs will always think the grass is always greener!
- Why do swimmers want to join your programme?
- The coaches reputation, the training times, the training space, the facility, convenience for them?
- Do new programmes attract people with issues in their current programmes?

- The 4 questions that I asked swimmers who wanted to join my new programme were as follows;
 - What are your goals?
 - What are you prepared to do in order to achieve your goals?
 - What do you expect from me as your coach?
 - What can you bring to the programme?
- Today I will ask the same questions to the swimmers who we are considering for a place on our National programmes in Great Britain

- Have a structural programme that runs through the whole programme.
 - » What squads you have?
 - » Entry criteria?
 - » Hours available for training on land and in the water etc.
- If you work or move within an established programme you should avoid making too many changes at once? Or should you?
- Remember the following,
 - The first 12 months in any job people come to you for advice.
 - The second 12 months they let you get on with the job following the changes you have made.
 - In the third 12 months the committee/board are now experts and they tell you what you should be doing.

- Have a 4-year plan and goals, that one-year is added to each season.
- - Work out where you want to go and how you are going to get there.
- - Each squad needs it's own goals and objectives.
- - Each squad needs its own seasonal plan – based on the needs of the athletes not on the needs and desires of the coach, club board/committee or parents.

Participation vs Performance



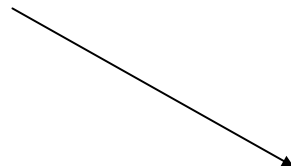
We need to bridge the gap between Participation and High Performance

Participation to Competitive

**8 Hours a week
is participation, fun, involvement
this is not competition swimming and
never produces a result**



**10-12 hours a week, at or beyond maturation
Too much to be fun and
They are not doing enough for a
competitive result – swimmers here never feel good and
become frustrated ‘The Twilight Zone’**



**Competitive
18 – 24 hours a week
you are in competitive swimming**

Does the following happen?

Smaller Clubs to pass on talent

The coach is ultimately responsible if they keep an athlete in the twilight zone and do not pass them on

If you have an athlete at maturation doing 10 hours a week what should you do?

Anything you want it will make no difference

Preparing Coaches for National Teams

- 1 All coaches need to understand that they are part of a team and whilst it is important they have the ability to prepare their athletes, they have to understand some things are controllable and others things are uncontrollable. Therefore focus on the controllable issues with their athletes.
- 2 Contact all swimmers home coaches and talk to them about their preparation prior to the meet, following their selection to the team.
- 3 Prepare individual athlete plans for each swimmer they are looking after into the championships. These plans to cover preparation pre championships, at the meet and when they have finished competing i.e. training in preparing for the next meet.

- 4 Find out, understand and accept team protocols
- 5 Be positive and accept team staff decisions as a united group for the team
- 6 Leave your personal and club team issues at home when you join the team
- 7 Know the whereabouts of all the athletes you are preparing on the team at all times
- 8 Call the home coaches of the athletes you are looking after every 2 to 3 days, or as necessary

- 9 Work with all team staff for the benefit of the team, whilst maintaining each individual athletes preparation.
- 10 Give each swimmer in your group the individual attention that they require, stagger warm ups etc
- 11 Be active on pool deck, coach the swimmers, do not instruct them. Chatting to other international coaches is fine, but not around the pool deck, if you have no reason to be on deck, go elsewhere.
- 12 If you see an issue address it, don't leave it for someone else.

13 Issues at coaches meetings stay in the meetings, present the united front to the team. Some issues at these meetings are just for the meetings.

14 Don't use negative motivation with the athletes using another staff member.

15 When you are on the national team, you are the national body.

16 Think of strategies to help swimmers attain fast heat swims i.e. wake up swims, walks showers etc

- 17 Be prepared to coach relay teams and work with other staff on their preparation, as well as preparing individuals
- 18 Conduct 3 relay practice sessions that are videoed and reviewed prior to the championships commencing
- 19 Coaches to deliver all team information to the athletes they are responsible for in a positive manner.
- 20 Understand that feedback you receive about your coaching is always a gift, even if you don't like it!

- 21 Know everything you can about the athletes you are working with, likes, dislikes, race strategy, manner etc
- 22 Get the race analysis of the athletes and review with them, look at the videos of the race etc
- 23 Swim downs, monitor the athletes, and ensure they do what is needed, this is the coaches responsibility
- 24 Talk to the athletes following the race, 3 or 4 quick fire questions before justifications set in

25 Communication with the athlete is all-important, they will decide in 3 training sessions of they,

- i) like and respect the coach – ideal
- ii) respect the coach, but don't really like – most can handle this one
- iii) like but don't respect the coach – not good at all
- iv) don't like or respect – what can you say to this?

26 Are you prepared to buy into all team protocols

27 Junior teams are about preparing for the senior team, the only end destination is the Olympic Games

Thoughts

- There is no such thing as an age group/youth 50m specialist!
- 200m events should be the priority for all age/youth athletes.
- Age groupers should focus on the 200/400IM as a priority.
- There should be no early specialisation for age group swimmers.
- 24 week Macro training cycles for age groupers and 15 week Macro training cycles for youth swimmers are the way to split your season.
- Are you as a coach committed or involved and do you know the difference.
- 1st September means swim training commences.
- Train through Christmas and New Year, no breaks.

Technique

- Technical Skill should be a priority.
- Key points should be known for all strokes

Butterfly

- The hands and arms accelerate through the final extension of the push phase of the stroke.
- When the hands and arms leave the water for the recovery phase the legs recover upwards ready to commence the downward kicking action.
- The face returns to the water slightly ahead of the entry of the hands to enable the smooth flow of the stroke as the hands go forward of the shoulders

- Butterfly Checklist
- 1. The swimmer should work on controlled frequency breathing patterns.
- 2. Holding the stroke length is of vital importance in Butterfly. (distance per stroke).
- 3. Always insist that the swimmer touches the wall with two hands at all times in training both on turns and on finishes.

Butterfly Checklist continued

- 4. The swimmer should have an ability to turn both ways (left and right).
- 5. The Butterfly swimmer should perform Butterfly kick sets so they are specific.
- 6. The coach should plan to include Butterfly training on all types of sets i.e. sprint and endurance sets.

- 7. Short rest sets can be utilized for endurance Butterfly sessions.
- 8. Some endurance Butterfly sets can also be done using a combination of full stroke with two, two, four or three, three, three Butterfly drills being incorporated into the set.
- 9 Butterfly kicks are practiced for speed in preference to distance and the differences in time should be compared between kicking on the front and the side.
- 10. Minimum maximum drill

Backstroke

- Backstroke is core stroke (simple skills)
Promote natural trunk rotation and early application of applied **force**
- Straight arm recovery
- Breathe on need
- Core strength and back strength
- Stable head position (only stroke with this)

Backstroke

- In Backstroke the importance of trunk rotation is crucial in order to allow the swimmer to have a deep catch at the commencement of the pull, with the palm facing downwards.
- The straight arm recovery and early application of force in the deep catch makes Backstroke a great stroke to teach in order to help with the development of the other strokes: many swimmers commenced their swimming careers as Backstroke swimmers before eventually specializing in other strokes.

- Another advantage with Backstroke is that breathing on need can take place--this, along with no movement of the head, means that breathing has no effect on the stroke mechanics.
- With no interruption to the body position, due to no variation in the head position, this ensures that the early application of force in the catch can be attained.
- Aerobic development can be done on Backstroke and it is a great stroke to offset all the work that is done on Freestyle.

Backstroke

- Entry ?
- Exit of the hand
- 6 beat kick

- Backstroke Checklist
- 1. The swimmers should start the first stroke on alternate arms in Backstroke sets to develop turning skills on both arms.
- 2. The swimmers should know how many Butterfly kicks to do off the start and turns. They should have these kicks from the start and the turns timed frequently at workouts in warm ups, main sets and swim downs.
- 3. Backstrokers should swim up to one half of weekly volume on Backstroke. This can include swim, drill, kick and pull.

- 4. Ensure the Backstroke flags are up for every session and are the correct distance from the wall (five meters).
- 5. The Backstroke swimmer needs to do more Backstroke kick and kicking drills than the Freestyle swimmer. The Backstroke swimmer should use a continuous six beat kick ensuring the knees stay under the water. A common fault is the pedaling action in the knees; this can be corrected by kicking with short fins or sneakers

Breaststroke

- Breaststroke is the least efficient of all the competitive swimming strokes, however it requires more natural skills due to the ankle and lower leg rotation that is required. The stroke requires greater amounts of skill training and coaching supervision during workouts.
- The value of the kick is essential to the efficiency of the stroke. The swimmer's flexibility, core strength, and individual size will play a part in determining the stroke technique of the swimmer.
- This stroke will probably be performed less on full stroke than the others and more work on drills, kick and pull would be done in training.

Breaststroke

- Sweep
- Hand Position
- Down, angle or up at the end of the in sweep

- 1. Allow Breaststrokers to warm up the legs by swim/kicking slowly at first, then build into a normal pace on training sets.
- 2. The amount of Breaststroke swum in each session should increase throughout the season. If the swimmer does experience knee pain switch strokes or change to a Breaststroke pull set or perform Breaststroke with a butterfly kick.

Breaststroke

- 3. Pulling drills are important for the Breaststroke swimmer, check whether palms are pitched at the correct angle i.e. thumbs down with the little fingers up on the outstroke.
- 4. Stroke drills for are essential for correct timing in Breaststroke.

- 5. The swimmer should finish every kick with accelerating foot speed and ensure the legs finish straight with the toes pointed, to maximize streamlining. The swimmer should not turn the feet out for the propulsive until the legs/feet are recovered fully.
- 6. The arm stroke insweep should move the body upwards and forwards, not upwards and backwards.

Front crawl

- When the swimmer commences in squad training they should have the basics well in place. Freestyle is traditionally the most used stroke for aerobic development and can therefore develop the most faults.

FC

- As a general rule the swimmer should have a balanced stroke that has sufficient trunk rotation to allow them to attain maximum distance per stroke and an early catch leading into a high stable elbow position underwater. This will result in maximum application of force when the stroke is fully rotated with minimum resistance.

Kick Training

- There is a belief that kick training is under used or under valued. These are some things to consider for your programme.
- Does your swimmer know their time for 100m-200m-300m-400m kick best times?
- Special attention should be placed on the percentage of butterfly kick (the 5th stroke) done during the total weekly volume, and whether done underwater and timed. Ensure that kick during swim downs is controlled. 50% of kick on butterfly, and then 50% of the butterfly kick should be done on the front and 50% on their back. Side kicking should also be considered. This is all important for Butterfly and Backstroke swimmers. Kicking Butterfly with a board, if done correctly, can be vital in strengthening the athlete's lower back.
- All kick in training should be timed and recorded.
- Sometimes you should consider finishing the workout with a kick challenge and then they can get out, i.e.:
 - 10 x 100m on 1.30s hold 1.25s or
 - 15 x 100m on 1.45s hold 1.35s or
 - 20 x 100m on 2.00s hold 1.45s
- Is 15% to 20% of your weekly volume on kick and do you use challenge kick sets?
- How often do you see a swimmer injured and then they have to train on kick only. They then have great improvements, when they are over the injury.
- A swimmer improving their kick can make big gains.
- Practise Front Crawl and Backstroke turns using a kickboard with either hands doing a full somersault on the wall.

Pull Training

- What is resistance pull? i.e. Band only for Freestyle and Backstroke, band and pull buoy for Breaststroke and Butterfly.
- Do your Breaststroke swimmers do Breaststroke (resistance) pull each week and monitor the distance covered? Breaststroke swimmers need to do Breaststroke pull to ensure upper body development occurs.
- Use the following with your swimmers 50m and 100m Band only pull timed and see if the differential gets smaller. This can be used to see if the swimmers pull is improving
- i.e. 35 seconds and 1:20 differential = 10
- 35 seconds and 1:18 differential = 8
- Therefore 2-second improvement.

Turns

- Turning drills on all strokes should be practised.
- Slow motion swimming with perfect skill, then an explosive turn for Backstroke and Freestyle, last hand enter to feet on wall in under 1 sec. The slower the swim the greater the skill demand and the faster the actual turn.
- Turning, hands to feet target of under 1 second for Breaststroke and Butterfly. This can also be used for BF to BK, BK to BR and BR to FS turns for medley swimmers.
- Work on turns under fatigue and at racing stroke rates? Time 7.5m in or 5m in and 10m out during training repeats, under pressure.
- 3 Breaststroke and Butterfly mid pool turns in 3 seconds or faster should be attained, before the following progression is performed.
- Swimmer faces away from the wall, and is sculling on the spot under the backstroke flags for each of the following steps in the progression, this progression is for use on Butterfly and Backstroke.
- One mid pool turn to start, then one turns at wall and the head passing the 5m flags in 7 seconds or faster.
- One mid pool turn to start, then three turns at wall and the head passing the 5m flags in 9 seconds or faster.
- Three mid pool turns to start, then one turn at wall and the head passing the 5m flags in 9 seconds or faster.
- Three mid pool turns to start, then three turns at wall and the head passing the 5m flags in 12 seconds or faster.
- How often do you insist on the swimmer performing the fly kicks underwater, following turns on training repeats in training that are expected in racing.

Starts

- Are starting blocks used in every workout? If not why not?
- Start each new set and session with a racing dive (Blocks).
- Track starts vs. grab starts which do you use and why?
- Are leg, back, and hip flexibility an issue for the athlete in the Mark position?
- Do the athletes use their arms to help propel themselves forward?
- Does the athlete maintain a rigid line upon entering the water?
- Do the athletes do the start they've always done or do they experiment with different hand/feet positions and opposites stances?
- Is vertical jumping and reaction drills part of the dryland program?
- Does the athlete sit on their back foot before starting?
- Practice in pairs and race to the 15m mark in the pool, alternate who initiates the start and the second swimmer reacts to their movement and aims to beat them into the water.

Relays

- Relay takeovers should be practised each week.
- Can your swimmer do individual heat, relay heat, individual final and relay final in competition? Have you prepared them? It is the coaches responsibility to prepare swimmers in the club/training programme so that they can achieve the above progression on the National team. Is this done in the club programme and at Nationals?
- Swimmers in general should be between 0.5 and 0.75 seconds faster in the relay performance than they are in the individual event.
- Learning relay takeovers the opposite way around i.e. break, break by a little less, break by a little less, and get them right, is a good way to learn. It is good to practice the extremes from very safe to very unsafe in a similar manner.
- Relay practice using 4 corners of the pool, 1 swimmer swims to a finish and 3 swimmers do a takeover, rotate through the group of 4.
- Does your club programme have a regular relay competition?
- Does the athlete have the ability to execute a fluid swing start; a one-foot step up start and a two-foot step up start? The ability to do all 3 correctly will bring confidence to the swimmer with the one they are most comfortable using.
- Does the athletes jump co-ordinate with the timing and momentum of the arm swing?
- Have you ever watched relay starts, when a takeover pad is in, at the major championships? Then write down what you think is the athletes takeover time, then check against the results of the meet and see how well you judge a takeover time.
- Does the breakout following the relay takeover hold the speed of the entry.
- What is a good takeover time? When practicing for relay takeovers as a general rule the swimmers will aim at a takeover time of between 0.10 and 0.20. Some swimmers can consistently be between 0.00 and 0.10, however it is suggested their accuracy be confirmed by using electronic timing pads in the training situation.

Talent and Skill

- It must always be noted that the experience of the coach or teacher must always be in advance of the talent of the athlete, so that the taught skills of the athlete will be in advance of that athlete's talent, but responsive to the coach's experience and knowledge. Quality of exposure to any given stimulus is far more productive and successful than extended time infrequently at exposure (i.e. it is better to practice 9 X 1 hour sessions than 4 X 2 hour sessions).
- Talent is a God-given hereditary skill. In nearly all cases where the skill of the athlete is learnt and developed over an extended period by the teacher and coach combination as applied to that athlete, ultimate potential can be realised. As a golden rule I always look to see that I have taught the athlete skills and they are in advance of the athlete's talent. I have witnessed very few coaches who can consistently achieve this high standard. This ensures that when the talent of one athlete to another is equal, then it is the taught skills that are not hereditary that will make up the difference.
- Formula 1 racing cars do not come off an assembly line, but are individually built. It is the benefits of these individually built racing cars where the technology is delivered downwards to normal cars used by families and individuals. Quality skills are learnt and taught by great teachers working within a systematic progression.

Work Ethos

- The second area that must be worked upon is the athlete work-ethos. The question here is, is the athlete able to work and willing to work. Some are able but unwilling, and this can be influenced and changed and redirected by great coaching.
- However, conversely some are willing to work, but are unable, mainly due to illness or injury or a breakdown in the immune system, relative to workloads. This is where the self-management, lifestyle and sports science influences become important to the final outcome of the athletes optimum performance.
- The ideal situation that everybody searches for is the combination where the athlete is both able to and willing to work at their talent levels or higher. This is a rare find in any given athlete, and one which all coaches search and look for and in some cases prioritise in my opinion above both talent (skill) and mental strength. Whilst this is rewarding, it can result in failing under pressure and under-achievement.
- Coaches recognise hard working athletes and sometimes fail to develop skills and mental strength. Where the programme has a fully integrated multi-dimensional system that is consistent, sustainable, repeatable and quality controlled, work ethos can be taught and developed in support of talent, skill and mental strength.

- 6 x 6 x 6 Teaching model:
 - That is the skill is practised six times. This must occur with the teacher/coach possessing varying teaching skills based on and combining both visual and audio teaching skills.
 - You as a teacher/coach have six ways to teach and achieve the same outcome
 - Practise each way for no longer than six minutes
 - Vary the presentation
- Teach butterfly initially with arms only over short periods and limited breathing.
- Teach 25m pools with early swimming development individual medley skills of:
 - 8 strokes butterfly
 - 12 strokes backstroke (single strokes)
 - 8 strokes breaststroke
 - 12 strokes front crawl (single strokes)

"Racing simulation Camp "

- One week, campus environment, lots of walking. The focus is to simulate the living and competition environment of a sports village. The camp will follow a competition program for the duration of the camp. The focus will be to put the swimmers through a heat, semi final and final type situation whilst maintaining a full training programme.
- The purpose of the camp is to demonstrate how competition goals can be achieved whilst maintaining long-term training requirements. This is a situation commonly experienced by National teams during long competition tours like world cup competitions. During this camp the swimmers will be divided up into sub groups based upon their competition events and their ability to recover and perform consistently therefore testing training strategies against competition performance.

"Racing simulation Camp "

- 1 day media training. Involvement on poolside sessions.
- Muscular skeletal screening conducted.
- Nutrition advice
- Mock drug test conducted following one of the swim sessions.
- Dry land training daily, Swiss ball, medicine ball and pilates - 5 days
- Referee and Starter to be invited in to this camp to conduct the racing simulation trials.
- Competition analysis conducted on the swimmers racing situation on selected athletes.
- Feedback session on performance modeling.
- Explanation of availability forms for competitions.
- Youth camps allow our top youth swimmers to practice the skills learnt at earlier stages of their careers and prepare themselves for National open representation.

Training Camp Equipment

Full training kit is required for poolside, this is the swimmers responsibility to bring to each camp and have available,

- Kickboard
- Pull Buoy
- Hand Paddles
- Finger Paddles
- Laminated heart rate/velocity curve from test set
- Race Model – laminated for poolside
- Hear Rate Monitor
- Plus anything else as advised pre camp

The home coach needs to educate their swimmers in,

- stroke rate,
- stroke count,
- using and understanding heart rate training levels,
- using a heart rate monitor
- streamlining,
- turns,
- finishes,
- relay takeovers

- IM swimming with Age Group Swimmers – all round development
- 50% of aerobic base on backstroke important
- 50% of kick on butterfly on both front and back
- Underwater butterfly kick 15m in 7 seconds or faster, every swimmer in GB can improve on this
- Start each new training section with a racing dive

- What is “fast”? Do you have a formula for speed work? Do you use speed charts or goal time i.e. end of season lock in speeds. How do you measure fast? In this same situation, what is “slow” aerobic? A speed can be evaluated for age group/youth swimmers based on 40 BBM and for seniors or white-fibred or heavily muscled athletes based on 50-60 BBM.
- Do all swimmers have their own goal lock-in speeds, for race speed and race pace training? Are these documented and recorded for all athletes with stroke rates and stroke length?
- Can your athletes swim at world record lock in speeds? Find a way to achieve i.e. assisted fins, tether, bands, shorter distance, longer rest etc.
- Repeats at World Record pace, goal pace and current PB pace on varying distances should be included in the weekly training plans, and example of this can be as follows,
 - 20m repeats at World Record pace.
 - 30m repeats at end of season goal pace.
 - 40m repeats at current PB pace
 - 20m, 30m, 40m repeats can also be used for kick and pull training.

- Minimum maximum back end speed values for training sets can be used and focus the swimmer on holding efficiency with speed at the back end of their race, i.e. If you have a swimmers race model, the last 50m can be used i.e. 29 seconds, 32 strokes, with a stroke rate of 42 for example. Therefore a well-documented end of season goal time race model should be available for each swimmer – laminated and on deck. The coach should always call total time of repeat and back-end speed in preference to front-end split.
- Back end speed sets should be done at specific time, stroke count and stroke rate. Remember compromise on rest, recovery and repeat distance not on the speed, stroke rate or stroke count.
- Use a specific number of strokes and record the distance travelled and time taken for another way to measure easy speed i.e. minimum/maximum. Then repeat for a specific number of repeats i.e. 10 or 12.

- Timed swims as against distance repeats (i.e.) 10 times 20 seconds repeats, measure the distance travelled, the stroke rate and the stroke length. Then repeat as number of strokes and measure distances. This should be done over an athlete's career so improvement can be measured from year to year of growth. This measurement can often act as an early indicator of plateauing performance barrier.
- Hold at least one stroke/event back prior to maturation so that any performance plateauing can be kick started and offset by exposure in competition to this under-exposed event, whilst resting the main event of the athlete.
- What drill progressions do you use? And why? Are the drills performed correctly?
- Do you use change of pace work in training, how often and done when?
- Do some sets where you have a physical goal in the morning and a mental goal in the afternoon.

- The psychology of a workout is important and should be understood by the coach, is it mentally stimulating and challenging. After the first set, grab their attention. Refocus them in the middle of long sets.
- There are many ways to grab their attention. Ask the athletes to call the skill to be worked on in each repeat or change the skill after each repeat and the coach should evaluate the skill being nominated before the next repeat. Athletes then buy into the workout.
- If recording heart rate/lactates always ask the swimmer to nominate their rate of perceived exertion value before offering the accurate measured reading and data. Swimmers should record their times so the coach can coach and not be a scribe. There should not be any need for pre-maturation athletes having lactates or heart rate exposure.